

- Description

This module manages a smart water pump system, ensuring that:

1. The system continuously monitors the current water flow rate (in liters per minute).
2. If the flow rate is below the minimum threshold, the pump speed is increased to ensure adequate water supply.
3. If the flow rate exceeds the maximum threshold, the pump speed is reduced to prevent excessive water flow.
4. If the flow rate is within the optimal range, the pump maintains its current speed.

This system helps optimize water distribution efficiency, prevent excessive consumption, and maintain stable water pressure.

- Specification

module SmartWaterPumpControl

var currentFlowRate, minFlowRate, maxFlowRate, pumpSpeed: real

process updateFlowRate(newFlowRate: real) returns (currentFlowRate: real, pumpSpeed: real)

functional scenario form:

currentFlowRate < minFlowRate && pumpSpeed = min(pumpSpeed * 1.2, 100.0), Pump speed increased.

||

currentFlowRate > maxFlowRate && pumpSpeed = max(pumpSpeed * 0.8, 20.0), Pump speed decreased.

||

currentFlowRate <= maxFlowRate && Pump speed unchanged.

end process

end module

- Java Code

```
package code;
public class c79_SmartWaterPump {
    private double minFlowRate;
    private double maxFlowRate;
    private double currentFlowRate;
    private double pumpSpeed;
    public c79_SmartWaterPump(double minFlowRate, double maxFlowRate, double initialFlowRate, double initialPumpSpeed) {
        this.minFlowRate = minFlowRate;
        this.maxFlowRate = maxFlowRate;
        this.currentFlowRate = initialFlowRate;
        this.pumpSpeed = initialPumpSpeed;
    }
    public void updateFlowRate(double newFlowRate) {
        currentFlowRate = newFlowRate;
    }
}
```

```

        adjustPumpSpeed();
        System.out.printf("Current Flow Rate: %.2f L/min, Pump Speed: %.2f%%\n", currentFlowRate, pumpSpeed);
        if (currentFlowRate < minFlowRate) {
            System.out.println("ALERT: Low water flow detected. Increasing pump speed.");
        } else if (currentFlowRate > maxFlowRate) {
            System.out.println("ALERT: High water flow detected. Reducing pump speed.");
        } else {
            System.out.println("Flow rate is optimal. Pump speed remains constant.");
        }
    }
    private void adjustPumpSpeed() {
        if (currentFlowRate < minFlowRate) {
            pumpSpeed = Math.min(pumpSpeed * 1.2, 100.0);
        } else if (currentFlowRate > maxFlowRate) {
            pumpSpeed = Math.max(pumpSpeed * 0.8, 20.0);
        }
    }
    public double getPumpSpeed() {
        return pumpSpeed;
    }
    public double getCurrentFlowRate() {
        return currentFlowRate;
    }
    public static void main(String[] args) {
        c79_SmartWaterPump pump = new c79_SmartWaterPump(10.0, 50.0, 30.0, 50.0);

        // testcase-VT:
        //     pump.updateFlowRate(5.0);
        //     pump.updateFlowRate(55.0);
        //     pump.updateFlowRate(35.0);

        // testcase-FT:
        //     pump.updateFlowRate(-1.7976931348606807E308);
        //     pump.updateFlowRate(0.0);
        //     pump.updateFlowRate(-4.9E-324);

        // testcase-Z3:
        //     pump.updateFlowRate(5.0);
        //     pump.updateFlowRate(55.0);
        //     pump.updateFlowRate(35.0);
    }
}

```

```

// testcase-UVT:
    pump.updateFlowRate(5.0);
    pump.updateFlowRate(55.0);
    pump.updateFlowRate(35.0);
}
}

```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c79_SmartWaterPump.java	100.0 %	8	0	8
▼ c79_SmartWaterPump	100.0 %	8	0	8
main(String[])		0	0	0
c79_SmartWaterPump(double, c		0	0	0
adjustPumpSpeed()	100.0 %	4	0	4
updateFlowRate(double)	100.0 %	4	0	4

3. Vibration testing instruction coverage: 94.8%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	110	20,987	21,097
▼ src	0.5 %	110	20,987	21,097
▼ code	0.5 %	110	20,987	21,097
▼ c79_SmartWaterPump.java	94.8 %	110	6	116
▼ c79_SmartWaterPump	94.8 %	110	6	116
main(String[])	100.0 %	18	0	18
c79_SmartWaterPump(double, c	100.0 %	15	0	15
adjustPumpSpeed()	100.0 %	30	0	30
updateFlowRate(double)	100.0 %	47	0	47

4. Fuzz testing branch coverage: 25%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	840	842
▼ src	0.2 %	2	840	842
▼ code	0.2 %	2	840	842
▼ c79_SmartWaterPump.java	25.0 %	2	6	8
▼ c79_SmartWaterPump	25.0 %	2	6	8
main(String[])		0	0	0
c79_SmartWaterPump(double, c		0	0	0
adjustPumpSpeed()	25.0 %	1	3	4
updateFlowRate(double)	25.0 %	1	3	4

5. Fuzz testing instruction coverage: 71.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	83	21,014	21,097
▼ src	0.4 %	83	21,014	21,097
▼ code	0.4 %	83	21,014	21,097
▼ c79_SmartWaterPump.java	71.6 %	83	33	116
▼ c79_SmartWaterPump	71.6 %	83	33	116
main(String[])	100.0 %	18	0	18
c79_SmartWaterPump(double, c	100.0 %	15	0	15
adjustPumpSpeed()	53.3 %	16	14	30
updateFlowRate(double)	72.3 %	34	13	47

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c79_SmartWaterPump.java	100.0 %	8	0	8
▼ c79_SmartWaterPump	100.0 %	8	0	8
main(String[])		0	0	0
c79_SmartWaterPump(double, c		0	0	0
adjustPumpSpeed()	100.0 %	4	0	4
updateFlowRate(double)	100.0 %	4	0	4

7. Z3 instruction coverage: 94.8%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	110	20,987	21,097
▼ src	0.5 %	110	20,987	21,097
▼ code	0.5 %	110	20,987	21,097
▼ c79_SmartWaterPump.java	94.8 %	110	6	116
▼ c79_SmartWaterPump	94.8 %	110	6	116
main(String[])	100.0 %	18	0	18
c79_SmartWaterPump(double, c	100.0 %	15	0	15
adjustPumpSpeed()	100.0 %	30	0	30
updateFlowRate(double)	100.0 %	47	0	47

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c79_SmartWaterPump.java	100.0 %	8	0	8
▼ c79_SmartWaterPump	100.0 %	8	0	8
main(String[])		0	0	0
c79_SmartWaterPump(double, c		0	0	0
adjustPumpSpeed()	100.0 %	4	0	4
updateFlowRate(double)	100.0 %	4	0	4

9. Unimproved V-method instruction coverage: 94.8%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	110	20,987	21,097
▼ src	0.5 %	110	20,987	21,097
▼ code	0.5 %	110	20,987	21,097
▼ c79_SmartWaterPump.java	94.8 %	110	6	116
▼ c79_SmartWaterPump	94.8 %	110	6	116
main(String[])	100.0 %	18	0	18
c79_SmartWaterPump(double, c	100.0 %	15	0	15
adjustPumpSpeed()	100.0 %	30	0	30
updateFlowRate(double)	100.0 %	47	0	47